

Tugas 1 Aljabar linear

Nama : Yudo Sandewo

Nim : 0529 685 24

1.1 Diketahui :

$$A = \begin{pmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

tentukan $1 * BA - B^T A^T = \begin{pmatrix} 5 & 6 & 9 \\ -6 & -3 & -6 \\ -15 & 0 & 1 \end{pmatrix}$

2.1 Diketahui Persamaan matriks

$$\begin{pmatrix} 5 & 4 \\ 1 & 8 \end{pmatrix} - \begin{pmatrix} 2 & 5 \\ 4 & 6 \end{pmatrix} X = \begin{pmatrix} 3 & -1 \\ -3 & 2 \end{pmatrix}$$

maka $\begin{pmatrix} 2 & 5 \\ 4 & 6 \end{pmatrix} X = \begin{pmatrix} 5 & 4 \\ 1 & 8 \end{pmatrix} - \begin{pmatrix} 3 & -1 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 4 & 6 \end{pmatrix}$

Karena Koefisien Sama. $X = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

3.1 Ubah matriks ke eselon ~~matrix~~ melalui operasi baris

diketahui $M = \begin{pmatrix} 2 & 4 & -2 & 6 \\ 1 & 3 & 1 & 4 \\ 3 & 7 & 5 & 8 \end{pmatrix}$

$$R_2 \leftarrow R_2 - \frac{1}{2} R_1, \quad R_3 \leftarrow R_3 - \frac{3}{2} R_1, \quad R_3 \leftarrow R_3 - R_2$$

eselon = $\begin{pmatrix} 2 & 4 & -2 & 6 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 6 & -2 \end{pmatrix}$

NO

DATE

4.1 diketahui :

$$A = \begin{pmatrix} 5 & 9 \\ 7 & 3 \end{pmatrix}$$

Jika $A - kI$ Singular. maka $\det(A - kI) = 0$

$$\det(A - kI) = (k - 12)(k + 4) = 0$$

Jadi $k = -4$ atau $k = 12$

5.1 Gauss Jordan Pada

$$P = \begin{pmatrix} 3 & 1 & -2 \\ 5 & -2 & -3 \\ 2 & 2 & 3 \end{pmatrix}$$

$$\text{REF Gauss Jordan} = I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$